

RONALDO MENEZES

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EDUCATION

University of York, England, UK

DPhil in Computer Science, 2000

Thesis: *Resource Management in Open Distributed Tuple Space Systems*

Advisor: Alan Wood

State University of Campinas, Brazil

MSc in Computer Science (Thesis Option), 1995

Thesis: *A Study on Models of Parallel Computation*

Advisor: João Carlos Setubal

University of Fortaleza, Brazil

BSc in Computer Science, 1992

Scholarships

- PhD Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) August 1995 – August 1999
- MSc Scholarship (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior, Brazil) January 1993 – December 1994

RESEARCH INTERESTS

Network Science, Complex Systems, Human Mobility, Urban Systems, Data Science, Social Network Analysis, Self-Organisation, Swarm Intelligence, Biologically-Inspired Computing, Science of Science, Inequality Modelling.

SELECTED RESEARCH GRANTS

◇ *Rapid Development of Novel Optimisation Tools for Transport and Energy Networks*

PI: Ed Keedwell; Co-PI: Ronaldo Menezes

Funded by Innovate UK, 2021–2024, £166,580

◇ *Methods for the Identification of Robbery, Theft of Artworks*

PI: Douglas Ferreira; External Collaborator: Ronaldo Menezes

Funded by FAPERJ-Brazil, 2019–2022, R\$250,000 (£35,000 approx.)

◇ *Understanding Human Movement in the Healthcare Sector using AI*

PI: Jacq Christmas; Co-PI: Ronaldo Menezes

Funded by Innovate UK, 2020–2022, £166,580

◇ *Hot Spot Policing Randomized Control Trial in the City of Exeter*

PI: Ronaldo Menezes; Co-PIs: Marcos Oliveira, Jo Arundale

Funded by Policing Lab, 2020–2021, £29,728

- ◇ *Identification of Human Mobility Modes using Socio-Spatio-Temporal Predictive Models*
 PI: Gourab Ghoshal; Co-PIs: Ronaldo Menezes, Brooke Welles
 Funded by US Army Research Office, 2018–2021, US\$500,000
- ◇ *Establishment of IANE (Northeastern Center for Artificial Intelligence), part of the IARA project*
 PI: Fernando Buarque; External Collaborator: Ronaldo Menezes
 Funded by a Private Consortium of Industries and Government Agencies, 2020, R\$7,000,000 (£1,000,000 approx.)
- ◇ *Automated Model Build for Decarbonisation and Climate Resilience*
 PI: Ronaldo Menezes; Co-PIs: Federico Botta, Marcos Oliveira
 Funded by Innovate UK, 2021, £90,751
- ◇ *Development of App and Web Interface for Automated Anuran Recognition and Mapping*
 PI: Mark Bush; Co-PIs: Ronaldo Menezes, Eraldo Ribeiro
 Funded by NSF, 2012–2018, \$348,000
- ◇ *Understanding the Predictability Limits in Human Mobility and Social Tie Formation in the Absence of Complete Data*
 PIs: Gourab Ghoshal, Ronaldo Menezes
 Funded by Army Research Office, 2016, \$60,000
- ◇ *InToPerformance: Influence of Network Topology to the Performance of Wireless Sensor Networks*
 PI: Angelo Brayner and Ronaldo Menezes
 Funded by Funcap (Brazil), 2013–2014, \$19,384
- ◇ *Generating Networks from Government Websites*
 PI: Ronaldo Menezes
 Funded by Harvard University, 2013, \$13,834
- ◇ *Exploring the Use of Nature-Inspired Algorithms in Software Testing Tools*
 PI: Ronaldo Menezes
 Funded by CNPq (Brazil), 2005–2007, \$110,000
- ◇ *Proactive, Selective Behavior Filtering with Data Restoration: A New Technique for Runtime Neutralization of Malicious Mobile Code*
 PI: James Whittaker; Co-PIs: Ronaldo Menezes, Cem Kaner, Michael Thomason, Jens Gregor and Gerald Marin
 Funded by Office of Naval Research, 2001–2005, \$4,000,000

PUBLICATIONS

Books

- ◇ *Methods of Statistical Physics Applied to Seismology: A Study of Earthquakes from the Viewpoint of Complex Networks*
 D. Ferreira, A. Papa, R. Menezes. LAP LAMBERT Academic Publishing, 2016.

Books Edited

- ◇ *Complex Networks XVI*
 M. Macedo, A. Cardillo, W. Franco, A. Brayner, R. Menezes. Springer Nature. 2025
- ◇ *Complex Networks XV*
 F. Botta, M. Macedo, H. Barbosa, R. Menezes. Springer Nature. 2024
- ◇ *Complex Networks XIV*

- A. Teixeira, F. Botta, J. F. F. Mendes, R. Menezes, G. Mangioni. Springer Nature. 2023
- ◇ *Complex Networks XIII*
D. Pacheco, A. Teixeira, H. Barbosa, R. Menezes, G. Mangioni. Springer Nature. 2022
 - ◇ *Complex Networks XII*
A. Teixeira, D. Pacheco, M. Oliveira, H. Barbosa, B. Gonçalves, R. Menezes. Springer Nature. 2021
 - ◇ *Complex Networks XI*
H. Barbosa, J. Gomez-Gardenes, B. Gonçalves, G. Mangioni, R. Menezes, M. Oliveira. Springer Nature. 2020
 - ◇ *Proceedings of the Intelligent Data Engineering and Automated Learning–IDEAL 2019: 20th International Conference*
H. Yin, D. Camacho, P. Tino, A. J. Tallón-Ballesteros, R. Menezes, R. Allmendinger. Manchester, UK, November 14–16, 2019, Springer Nature.
 - ◇ *Complex Networks VIII*
B. Gonçalves, R. Menezes, R. Sinatra, V. Zlatic. Springer International Publishing. 2017
 - ◇ *Complex Networks VII*
H. Cherifi, B. Gonçalves, R. Menezes, R. Sinatra. Studies in Computational Intelligence, Vol. 644. 2016
 - ◇ *Complex Networks V*
P. Contucci, R. Menezes, A. Omicini, J. Poncela-Casasnovas. Studies in Computational Intelligence, Vol. 549. 2014
 - ◇ *Complex Networks IV*
R. Menezes, A. Evsukoff and M. C. González. Studies in Computational Intelligence, Vol. 424. 2013
 - ◇ *Complex Networks: Second International Workshop, CompleNet 2010*
L. F. Costa, A. Evsukoff, G. Mangioni and R. Menezes. Communications in Computer and Information Science, Vol. 116. 2011
 - ◇ *Complex Networks: Results of the 1st International Workshop on Complex Networks (CompleNet 2009)*
R. Menezes, S. Fortunato, G. Mangioni and V. Nicosia. Studies in Computational Intelligence, Vol. 207. 2009

BOOK CHAPTERS

- ◇ *Towards intelligent dynamics of an active transport system for biking*
A. Medaglia, M. Wilches-Mogollon, O. Sarmiento, F. Montes, L. Guzman, M. Sanchez-Silva, R. Menezes, D. Hidalgo, K. Parra, A. Useche, H. Ochoa-Montero, N. Rodríguez, L. Salamanca. In *Safer Complex Systems Case Studies*. Royal Academy of Engineering. May 2022.
- ◇ *Networks in Rural Sociology*
Ronaldo Menezes and Pramod Divakarmurthy. In *Encyclopedia on Social Network Analysis and Mining*, eds. Reda Alhajj and Jon Rokne. August 2014.
- ◇ *Simulating Crime Against Properties Using Swarm Intelligence and Social Networks*
V. Furtado, A. Melo, A. Coelho, R. Menezes and M. Belchior. *Artificial Crime Analysis Systems*. IGI Global, 2008. 300–318.
- ◇ *Scalability in Linda-like Coordination Systems*
R. Menezes, R. Tolksdorf and A. Wood. In *Coordination of Internet Agents: Models, Technologies and Applications*, eds. A. Omicini, F. Zambonelli, M. Klush and R. Tolksdorf. Springer-Verlag. 2001

UNDER REVIEW AND ONGOING

- ◇ *A shared IPTC topic space for cross-source topic modelling*
D. Iskakov, S. Gonçalves, M. Idiat, M. Vainstein, A. Villavicencio, R. Menezes, R. Wilkens. arXiv preprint arXiv:2606.26845. 2026
- ◇ *Systematic comparison of gender inequality in scientific rankings across disciplines*
A. M. Jaramillo, M. Macedo, M. Oliveira, F. Karimi, R. Menezes. arXiv preprint arXiv:2501.13061 (under review *Nature Human Behaviour*)
- ◇ *Super-Linear Growth and Rising Inequality in Online Social Communities: Insights from Reddit*
G. Machado, D. Pacheco, R. Menezes, G. Baxter. arXiv preprint arXiv:2503.02661 (under review *EPJ Data Science*)

JOURNAL

- ◇ *The parenthood effect in urban mobility*
M. Macedo, R. Menezes, A. Cardillo. npj Complexity 3, 23. 2026
- ◇ *Validating urban scaling laws through mobile phone data: A continental-scale analysis of Brazil's largest cities*
R. S. Alencar, F. L. Ribeiro, H. Samaniego, R. Menezes, A. G. Evsukoff. Computational Urban Science 6 (1), 21. 2026
- ◇ *Network analysis of cultural funding in Brazil: Uncovering regional differences and signatures*
D. Pacheco, G. Moraes, D. Pinheiro, J. Faustino, M. Oliveira, H. Menezes, R. Menezes. Applied Network Science 11 (1), 43. 2026
- ◇ *Natural language processing of biomedical text to map and prioritize protein–disease associations in HFP_{EF}*
C. Santana, C. Mukherjee, A. Quazi, R. Menezes, V. Filkov, D. Sigdel, H. Choi, I. Ebong, P. Sirish, N. R. Anderson, X. Wang, H. Ji, J. Han, B. Roshanravan, L. T. Izu, T. W. Smith, N. Chiamvimonvat, C. E. Clancy, M. Cadeiras, D. A. Liem. Computers in Biology and Medicine 205, 111599. 2026
- ◇ *Vector fields as a framework for modelling the mobility of commodities*
S. Farokhnejad, A. S. da Mata, M. Macedo, R. Menezes. PLoS One 21 (3), e0340109. 2026
- ◇ *Not all roads are created equal: network science shows some highways matter more for Brazil's connectivity*
J. Taveira, R. Menezes, F. Buarque. Applied Network Science 11 (1), 25. 2026
- ◇ *Violence, city size and geographical isolation in African cities*
R. Prieto-Curiel, R. Menezes. Nature Communications 16 (1), Article 10049. 2025
- ◇ *Beyond proximity: A review and framework to further understanding of greenspace accessibility in the x-minute city*
A. Schendl, J. M. Bullock, R. Menezes, S. Willcock. People and Nature 7 (7), 1491–1504. 2025
- ◇ *A data-driven supervised machine learning approach to estimating global ambient air pollution concentrations with associated prediction intervals*
L. J. Berrisford, H. Barbosa, R. Menezes. Royal Society Open Science 12 (7), 241288. 2025
- ◇ *The impact of the social exposome in cardiovascular health and disease*
K. El Mekkaoui, C. Mukherjee, C. Santana, H. Silva, A. Quazi, R. Menezes. The Journal of Precision Medicine: Health and Disease, 100015. 2025
- ◇ *A framework for scalable ambient air pollution concentration estimation*
L. J. Berrisford, L. S. Neal, H. J. Buttery, B. R. Evans, R. Menezes. Environmental Data Science, Volume 4, e17. 2025

- ◇ *Mobility and transit segregation in urban spaces*
N. Iyer, R. Menezes, H. Barbosa. *Environment and Planning B: Urban Analytics and City Science* 51 (7), 1496–1512. 2024
- ◇ *The role of transport systems in housing insecurity: a mobility-based analysis*
N. Iyer, R. Menezes, H. Barbosa. *EPJ Data Science* 13 (1), 49. 2024
- ◇ *Dynamic predictability and activity-location contexts in human mobility*
B. Poudyal, D. Pacheco, M. Oliveira, Z. Chen, H. S. Barbosa, R. Menezes, G. Ghoshal. *Royal Society Open Science* 11 (9), 240115. 2024
- ◇ *Dynamics of consumption of child sexual abuse material on the dark web*
P. Divakarmurthy, B. Cunha, J. Passoud, R. Menezes, M. Oliveira. *PLoS One* 19 (7), e0306516. 2024
- ◇ *Environmental Insights: Democratizing access to ambient air pollution data and predictive analytics with an open-source Python package*
L. J. Berrisford, R. Menezes. *Environmental Modelling & Software*, Volume 179, 106131. 2024
- ◇ *Estimating annual ambient air pollution using structural properties of road networks*
L. J. Berrisford, E. Ribeiro, R. Menezes. *Environment and Planning B: Urban Analytics and City Science* 51 (9), 2031–2054. 2024
- ◇ *The structure of segregation in co-authorship networks and its impact on scientific production*
A. M. Jaramillo, H. T. P. Williams, N. Perra, R. Menezes. *EPJ Data Science* 12 (1), 47. 2023
- ◇ *Transport and Mobility Segregation in Urban Spaces*
N. Iyer, R. Menezes, H. Barbosa. *International Journal of Population Data Science* 8 (3). 2023
- ◇ *COVID-19 is linked to changes in the time–space dimension of human mobility*
C. Santana, F. Botta, H. Barbosa, F. Privitera, R. Menezes, R. Di Clemente. *Nature Human Behaviour*, 1–11. 2023
- ◇ *Contrasting social and non-social sources of predictability in human mobility*
Z. Chen, S. Kelty, A. Evsukoff, B. F. Welles, J. P. Bagrow, R. Menezes, G. Ghoshal. *Nature Communications*, Volume 13, Article 1922. 2022
- ◇ *Beyond exploitation: Measuring the impact of local search in swarm-based memetic algorithms through the interactions of individuals in the population*
C. Santana, M. Oliveira, C. Bastos-Filho, R. Menezes. *Swarm and Evolutionary Computation*, 101040. 2022
- ◇ *Differences in the spatial landscape of urban mobility: gender and socioeconomic perspectives*
M. Macedo, L. Lotero, A. Cardillo, R. Menezes, H. Barbosa. *PLoS One*. <https://doi.org/10.1371/journal.pone.0260874>. 2022
- ◇ *Reaching to the top: The gender effect in highly-ranked academics in computer science*
A. Jaramillo, M. Macedo, R. Menezes. *Advances in Complex Systems*, Vol. 24, No. 03n04, 2150008. 2021
- ◇ *Breast cancer diagnosis using thermal image analysis: A data-driven approach based on swarm intelligence and supervised learning for optimized feature selection*
M. Macedo, M. Santana, W. P. dos Santos, R. Menezes, C. Bastos-Filho. *Applied Soft Computing*, 107533. 2021
- ◇ *Uncovering the social interaction network in swarm intelligence algorithms*
M. Oliveira, D. Pinheiro, M. Macedo, C. Bastos-Filho, R. Menezes. *Applied Network Science* 5, 1–20. 2020
- ◇ *Uncovering the differences and similarities between physical and virtual mobility*
S. Hazarie, H. Barbosa, A. Frank, R. Menezes, G. Ghoshal. *Journal of the Royal Society Interface*

- ◇ *A data-driven social network intervention for improving organ donation awareness among minorities: analysis and optimization of a cross-sectional study*
M. D. Murphy, D. Pinheiro, R. Iyengar, G. Lim, R. Menezes, M. Cadeiras. *Journal of Medical Internet Research* 22 (1), e14605. 2020
- ◇ *A data-driven network approach for characterization of political parties' ideology dynamics*
J. Faustino, H. Barbosa, E. Ribeiro, R. Menezes. *Applied Network Science* 4 (1), 48. 2019
- ◇ *An approach for thematic relevance analysis applied to textual contributions in discussion forums*
C. J. R. Machado, A. M. A. Maciel, R. L. Rodrigues, R. Menezes. *International Journal of Distance Education Technologies (IJDET)* 17 (3), 37–51. 2019
- ◇ *A data-driven framework for monitoring and improving organ donation awareness among minorities using optimal social network intervention*
D. Pinheiro, M. Murphy, R. Iyengar, J. Lazareff, J. Choi, R. Menezes, M. Cadeiras. *The Journal of Heart and Lung Transplantation* 38 (4), S27. 2019
- ◇ *Unveiling the interplay between the TLR4/MD2 complex and HSP70 in the human cardiovascular system: a computational approach*
A. A. de Oliveira, J. Faustino, M. E. de Lima, R. Menezes, K. P. Nunes. *International Journal of Molecular Sciences* 20 (13), 3121. 2019
- ◇ *Spatio-temporal variations in the urban rhythm: the travelling waves of crime*
M. Oliveira, E. Ribeiro, C. Bastos-Filho, R. Menezes. *EPJ Data Science* 7 (1), 29. 2018
- ◇ *Towards evidence of long-range correlations in shallow seismic activities*
D. Ferreira, J. Ribeiro, A. Papa, R. Menezes. *EPL (Europhysics Letters)* 121 (5), 58003. 2018
- ◇ *Human mobility: Models and applications*
H. Barbosa-Filho, M. Barthelémy, G. Ghoshal, C. R. James, M. Lenormand, T. Louail, R. Menezes, J. J. Ramasco, F. Simini, M. Tomasini. *Physics Reports* 734, 1–74. 2018
- ◇ *The scaling of crime concentration in cities*
M. Oliveira, C. Bastos-Filho, R. Menezes. *PLoS One* 12 (8), e0183110. 2017
- ◇ *On the effect of human mobility to the design of metropolitan mobile opportunistic networks of sensors*
M. Tomasini, B. Mahmood, F. Zambonelli, A. Brayner, R. Menezes. *Pervasive and Mobile Computing* 38, 215–232. 2017
- ◇ *The role of human relations and interactions in designing memory-related models for sensor networks*
B. Mahmood, R. Menezes. *Sensors & Transducers* 199 (4), 42. 2016
- ◇ *The effect of recency to human mobility*
H. Barbosa, F. B. Lima-Neto, A. Evsukoff, R. Menezes. *EPJ Data Science* 4:21. 2015
- ◇ *Social-driven information dissemination for mobile wireless sensor networks*
B. Mahmood, M. Tomasini, R. Menezes. *Sensors and Transducers Journal*, Volume 189, Issue 6. 2015
- ◇ *On the agreement between small-world-like OFC model and real earthquakes*
D. S. R. Ferreira, A. Papa, R. Menezes. *Physics Letters A*, Volume 379, Issue 7. 2015
- ◇ *Using network science to assess particle swarm optimizers*
M. Oliveira, C. J. A. Bastos-Filho, R. Menezes. *Social Network Analysis and Mining* 5:3. 2015
- ◇ *Using social media as a method of increasing organ donation*
M. Cadeiras, M. Soroya, S. Patel, C. Starling, G. Lim, K. Dong, R. Menezes, M. Deng. *Circulation. Suppl.* 2. 2014
- ◇ *Small world picture of worldwide seismic events*

- D. S. R. Ferreira, A. Papa, R. Menezes. *Physica A*, Volume 408, pp. 170–180. 2014
- ◇ *A self-organized approach for detecting communities in networks*
B. Collingsworth and R. Menezes. *Social Network Analysis and Mining* 4:169. 2014
 - ◇ *Understanding the spread of malicious mobile-phone programs and their damage potential*
P. Wang, M. C. González, R. Menezes, A.-L. Barabási. *International Journal of Information Security* 12 (5): 383–392. 2013
 - ◇ *The United States heart and lung transplant “social” network: insights from the UNOS database*
S. Venugopal, E. Stoner, M. Deng, R. Menezes, M. Cadeiras. *Journal of Heart and Lung Transplantation*, Vol. 32, Issue 4, Supplement, S159–S160. 2013
 - ◇ *Understanding organ transplantation in the USA using geographical social networks*
S. Venugopal, E. Stoner, M. Cadeiras, R. Menezes. *Social Network Analysis and Mining* 3 (3): 457–473. 2013
 - ◇ *A bio-inspired crime simulation model*
V. Furtado, A. Melo, A. L. V. Coelho, R. Menezes, R. Perrone. *Decision Support Systems* 48 (1): 282–292. 2009
 - ◇ *Knowledge tuples never die, they just fade away*
L. Nixon, R. Tolksdorf, R. Menezes, A. Wood. *Ubiquitous Computing and Communication Journal*, Vol. 3. 2008
 - ◇ *An adaptive in-network aggregation operator for query processing in wireless sensor networks*
A. Brayner, A. Lopes, D. Meira, R. Vasconcelos, R. Menezes. *Journal of Systems and Software*, Volume 81, Issue 3, pp. 328–342. 2008
 - ◇ *Intelligent systems at Florida Tech*
P. K. Chan, R. Menezes, D. Mitra, E. Ribeiro, M. Silaghi. *IEEE Intelligent Informatics Bulletin* 8 (1): 5–6. 2007
 - ◇ *Toward adaptive query processing in wireless sensor networks*
A. Brayner, A. Lopes, D. Meira, R. Vasconcelos, R. Menezes. *Signal Processing*, Volume 87, No. 12, pp. 2911–2933. 2007
 - ◇ *Case studies for self-organization in computer science*
M. Mamei, R. Menezes, R. Tolksdorf, F. Zambonelli. *Journal of Systems Architecture* 53 (8-9), pp. 443–460. 2006
 - ◇ *On coordination and its significance to distributed and multi-agent systems*
S. Ossowski and R. Menezes. *Concurrency and Computation: Practice and Experience*. 2005
 - ◇ *On the semantics of coordination models for distributed systems: The LogOp case study*
R. Menezes, A. Omicini, M. Viroli. *Electronic Notes in Theoretical Computer Science*, Volume 97, pp. 97–124. 2004
 - ◇ *Resource management in open Linda systems*
R. Menezes. *Concurrency and Computation: Practice and Experience*, Vol. 15, pp. 1233–1256. 2003
 - ◇ *Coordination in a content-addressable web*
R. Menezes, I. Merrick, A. Wood. *Autonomous Agents and Multi-Agent Systems Journal*, Vol. 2, No. 3, pp. 287–301. 1999

PEER-REVIEWED CONFERENCES AND WORKSHOPS

- ◇ *Network-centric analysis of memetic operators and communication topologies for swarm intelligence algorithms*
C. Silva, C. Santana, R. Menezes, C. Bastos-Filho. *International Conference on Complex Networks*, 27–40. 2025

- ◇ *Understanding the structure and resilience of the Brazilian federal road network through network science*
J. Taveira, F. B. L. Neto, R. Menezes. International Conference on Complex Networks, 107–121. 2025
- ◇ *The rephrased reality: Analysing sentiment shifts in LLM-rephrased text*
F. L. Richardson, A. Villavicencio, R. Menezes. Proceedings of the 40th ACM/SIGAPP Symposium on Applied Computing, 920–927. 2025
- ◇ *Academic networks and resilience: An education perspective*
P. Velasquez-Ferrufino, R. Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Network-based analysis of crossover operators in genetic algorithms*
R. Rosenbaum, R. Menezes, C. Santana. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Branching out: How academic genealogy networks shape academic groups*
G. Moraes, A. Brayner, R. Menezes. Complex Networks & Their Applications XIII. Istanbul, Turkey. 2024
- ◇ *Multidimensional analysis of the influence of socioeconomic and political indicators on the spread of COVID-19: A case study of Brazilian cities (2020–2024)*
R. Reis, A. Brayner, M. Ângelo, R. Menezes. Simpósio Brasileiro de Banco de Dados (SBBD), 394–405. 2024
- ◇ *Public transit inequality in the context of the built environment*
N. Iyer, R. Menezes, H. Barbosa. International Conference on Complex Networks, 203–218. 2024
- ◇ *Social interactions matter: Is grey wolf optimizer a particle swarm optimization variation?*
R. C. Lira, M. Macedo, H. V. Siqueira, R. Menezes, C. Bastos-Filho. International Conference on Complex Networks, 101–114. 2024
- ◇ *Nature vs. nurture in science: The effect of researchers segregation on papers' citation histories*
A. M. Jaramillo, F. Montes, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 141–154. 2023
- ◇ *Network entropy as a measure of socioeconomic segregation in residential and employment landscapes*
N. Iyer, R. Menezes, H. Barbosa. Proceedings of the 14th Conference on Complex Networks (CompleNet), 26–37. 2023
- ◇ *Academic mobility as a driver of productivity: a gender-centric approach*
M. Macedo, A. M. Jaramillo, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 120–131. 2023
- ◇ *Using vector fields in the modelling of movements as flows*
S. Farokhnejad, M. Oliveira, E. Ribeiro, R. Menezes. Proceedings of the 14th Conference on Complex Networks (CompleNet), 155–167. 2023
- ◇ *A data-driven approach to cattle epidemic modelling under uncertainty*
S. Farokhnejad, D. Cardoso, C. Rocha, A. S. da Mata, R. Menezes. Proceedings of the 13th Conference on Complex Networks (CompleNet). 2022
- ◇ *Networks of evolution: Modelling and deconstructing genetic algorithms using dynamic networks*
C. Santana, E. Keedwell, R. Menezes. Proceedings of the Genetic and Evolutionary Computation Conference. 2022
- ◇ *Modelling the social interactions in grey wolf optimizer*
R. C. Lira, M. Macedo, H. V. Siqueira, R. Menezes, C. Bastos-Filho. IEEE Latin American Conference on Computational Intelligence (LA-CCI). 2021

- ◇ *Fishing for interactions: A network science approach to modeling fish school search*
M. Macedo, L. Taw, N. Gurrapadi, R. C. Lira, D. Pinheiro, M. Oliveira, R. Menezes. Proceedings of the Genetic and Evolutionary Computation Conference, 40–48. 2021
- ◇ *An approach to assess swarm intelligence algorithms based on complex networks*
C. Santana, E. Keedwell, R. Menezes. Proceedings of the 2020 Genetic and Evolutionary Computation Conference, 31–39. 2020
- ◇ *A longitudinal analysis of vocabulary changes in social media*
H. Hamoodat, F. Aswad, E. Ribeiro, R. Menezes. Complex Networks XI, 212–221. 2020
- ◇ *Communities of human migration in social media: An experiment in social sensing*
F. Aswad, H. Hamoodat, E. Ribeiro, R. Menezes. Complex Networks XI, 222–232. 2020
- ◇ *Characterizing the dynamics of academic affiliations: A network science approach*
J. Faustino, N. Iyer, J. Mendonza, R. Menezes. Complex Networks XI, 393–404. 2020
- ◇ *Network-based delineation of health service areas: A comparative analysis of community detection algorithms*
D. Pinheiro, R. Hartman, E. Romero, R. Menezes, M. Cadeiras. Complex Networks XI, 359–370. 2020
- ◇ *Gender patterns of human mobility in Colombia: Reexamining Ravenstein’s laws of migration*
M. Macedo, L. Lotero, A. Cardillo, H. Barbosa, R. Menezes. Complex Networks XI, 269–281. 2020
- ◇ *Short text tagging using nested stochastic block model: A Yelp case study*
J. Bowllan, K. Cozart, S. M. M. Seyednezhad, A. Smith, R. Menezes. International Conference on Complex Networks and Their Applications, 822–833. 2019
- ◇ *Modelling the social interactions in ant colony optimization*
N. Gurrapadi, L. Taw, M. Macedo, M. Oliveira, D. Pinheiro, C. Bastos-Filho, R. Menezes. International Conference on Intelligent Data Engineering and Automated Learning, 216–222. 2019
- ◇ *Characterizing the social interactions in the artificial bee colony algorithm*
L. Taw, N. Gurrapadi, M. Macedo, M. Oliveira, D. Pinheiro, C. Bastos-Filho, R. Menezes. IEEE Congress on Evolutionary Computation (CEC), 1243–1250. 2019
- ◇ *Social media vocabulary reveals education attainment of populations*
H. Hamoodat, E. Ribeiro, R. Menezes. International Workshop on Complex Networks, 157–168. 2019
- ◇ *Entropy in network community as an indicator of language structure in emoji usage: A Twitter study across various thematic datasets*
R. Hartman, S. M. M. Seyednezhad, D. Pinheiro, J. Faustino, R. Menezes. International Conference on Complex Networks and Their Applications, 328–337. 2018
- ◇ *Author attribution using network motifs*
Y. Al Rozz, R. Menezes. International Workshop on Complex Networks, 199–207. 2018
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R. Menezes, R. Tolksdorf. Florida Institute of Technology, Department of Computer Sciences, CS-2003-04. 2003
- ◇ *On the use of Linda as a framework for distributed database systems*
M. Thirukonda, R. Menezes. Florida Institute of Technology, Department of Computer Sciences, CS-2002-02. 2002
- ◇ *Ligia: A Java based Linda-like run-time system with garbage collection of tuple spaces*
R. Menezes, A. Wood. University of York, Department of Computer Science, YCS 304. 1998
- ◇ *Parallel computing models and algorithm design (in Portuguese)*
R. Menezes, J. C. Setubal. Institute of Computing, State University of Campinas, Brazil. 1995

EXHIBITIONS

The Art of Networks II, Poland, January 2016

The Art of Networks II, Foosaner Art Museum, September–November 2015

The Art of Networks II, New York Hall of Science, March–May 2015

The Art of Networks, Foosaner Art Museum, March–April 2012

Analyse This: Digital Models of Swarm Intelligence, Ruth Funk Center for Textile Arts, September 2010

WORK EXPERIENCE

ACADEMIC

Computer Science Director of Global Development

University of Exeter, UK

September 2024 – Present

Professor of Data and Network Science

University of Exeter, UK

August 2018 – Present

External Examiner

University of Sussex, UK

January 2025 – December 2025

Visiting Professor

Federal University of Ceará, Brazil

October 2022 – 2024

External Examiner

University of Warwick, UK

January 2021 – 2024

Alan Turing Fellow

The Alan Turing Institute, UK

October 2021 – January 2024

Head of the Computer Science Department

University of Exeter, UK

August 2018 – August 2022

Professor in Computer Science

Florida Institute of Technology, USA

March 2016 – 2018

Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Computing Careers and Disciplines.

Joint appointments: School of Human-Centered Design, Innovation & Art

Director of the School of Computing

Florida Institute of Technology, USA

December 2016 – October 2017

Department Head of Computer Science

Florida Institute of Technology, USA

December 2015 – December 2016

Associate Professor in Computer Science

Florida Institute of Technology, USA

August 2006 – 2016

Teaching Activities: Social Network Analysis, Complex Networks, Algorithms and Data Structures, Swarm Intelligence, Introduction to Biologically Inspired Computing.

Visiting Researcher

Department of Physics, Northeastern University, CCNR (Barabasi Lab), Boston, USA January 2009 –

July 2009

Visiting Researcher

IVIA Ltda., Brazil

September 2005 – 2007

Assistant Professor in Computer Science

Florida Institute of Technology, USA

August 2000 – 2006

Teaching Activities: Java, Distributed Systems, Network Programming, Algorithms and Data Structures, Coordination Systems, Swarm Intelligence, Introduction to Biologically Inspired Computing.

Lecturer in Computer Science (Assistant Professor)

University of Ulster, UK

August 1999 – July 2000

Teaching Activities: Object Oriented Programming

Teaching Assistant

The University of York, UK

August 1996 – July 1999

Teaching Activities: Theory of Computation, Pascal and Java Programming.

Adjunct Faculty in Computer Science

Padre Anchieta University, Brazil

November 1994 – June 1995

Teaching Activities: Algorithms and Data Structures

INDUSTRIAL

Web Technology Consultant

8over8 Limited, UK

February 2000 – July 2000

Systems Consultant

Mota Machado Constructors, Brazil

January 1993 – June 1995

Computing Services Director

Fiutex Textile Industry, Brazil

July 1991 – February 1993

Programmer / System Analyst

Mota Machado Constructors, Brazil

October 1989 – July 1991

TEACHING ACTIVITIES

GRADUATE COURSES

Complex Networks

An advanced version of the Social Network Analysis course. Students produce a conference-quality research paper. The course includes heavier mathematical formalism and studies non-social networks (technological, biological, etc.)

Social Networks Analysis

A seminar-like course on computational approaches to social networks covering modelling, data collection, and structural analysis.

Introduction to Biologically Inspired Computing

An introductory course covering most areas of computer science inspired by biological systems.

Swarm Intelligence

An advanced course for senior graduate students. One of the first courses in the area offered worldwide; material has been used at EPFL (Switzerland) and JPL/Caltech (USA).

Coordination Systems

An advanced course on the theory and practice of coordination. The first course in the world on this topic, covering computation and coordination as distinct components of distributed systems.

Distributed Systems

Standard course on distributed systems as a core area of computer science.

Network Programming

A project-based course where students write programs using computer network concepts.

UNDERGRADUATE COURSES

Fundamentals of Software Development II

A freshman programming course reorganised to encourage diligent study with constant feedback. Includes 4 exams, 10 pop quizzes, 2 semester projects, and 15 lab exercises.

Algorithms and Data Structures

A core CS curriculum course emphasising lab practice and continuous feedback.

SELECTED INVITED AND KEYNOTE TALKS

Finding the ‘hidden’ dimensions of inequalities Enredando 2026, Viçosa, Minas Gerais, Brazil. July 2026

Recipe/Food Networks as Windows into Inequality: A Call for Computational Analysis of Eating Behaviours. First International Workshop of Scientific and Computational Gastronomy. Parma, Italy. September 2025

Violence, City Size and Geographical Isolation in African Cities. LANET. Punta del Este, Uruguay. August 2025

Random Thoughts about Causality and Mobility. European Mobility Symposium. Pisa, Italy. November 2024

Uncovering the differences and similarities between physical and virtual mobility. Lipari School of Complex Systems. Italy. July 2023

Gender Inequality in Scientific Production. LANET. Cusco, Peru. August 2023

Disaggregating Data: A Case of Gender Differences in Collaboration Networks and Urban Human Mobility. Data Science Africa. Arusha, Tanzania. July 2022

Differences in the spatial landscape of urban mobility: Gender and socioeconomic perspectives. Alan Turing Institute. Urban Analytics. Leeds, UK. 2021

Network Science as a Framework for Understanding Population Health. One Health One Planet Meeting. Lavras, Minas Gerais, Brazil. 2021

Using Data Analytics to Understand Human Mobility Patterns in Cities. Analytics Forum. Colombia. 2020

The Next Big Thing in Big Data. CSoNet. Ho Chi Minh City. Vietnam. 2019

On the Structure, Concentration and Dynamics of Urban Crime. LA-CCI. Guayaquil, Ecuador. 2019

On the Structure, Concentration and Dynamics of Urban Crime. MARAMI. Dijon. France. 2019

On the Scale, Concentration and Dynamics of Crime. Urban Analytics Forum. Alan Turing Institute. Bristol. 2019

On the Scale, Concentration and Dynamics of Crime. Cambridge Networks Day. 2019

On the Scale, Concentration and Dynamics of Crime. LANET. Cartagena, Colombia. 2019

On the Scale, Concentration and Dynamics of Crime. Lipari School of Complex Systems. Italy. July 2019

Human Dynamics and Landscape Decisions: Why People and Urban Systems Matter. Isaac Newton Institute. University of Cambridge, UK. July 2019

On the Scale, Concentration and Dynamics of Crime. University of Aveiro, Portugal. July 2019

On the Scale, Concentration and Dynamics of Crime. Alan Turing Institute, London. June 2019

Crime Concentration in Cities. IFISC. Palma, Spain. 2019

Self-Organisation in Computer Science. Christmas Lecture, University of Exeter, UK. 2018

Spatio-Temporal Regularities in Crime Dynamics. Oxford Mathematics Institute, UK. 2018

Human Mobility. LA-CCI. Arequipa, Peru. 2017

The Scaling of Crime Concentration in Cities. LANET. Puebla, Mexico. 2017

Assessing Swarm System using Network Sciences. NetSci-X 2015. Rio de Janeiro. 2015

Human Mobility: Social Relations Applied to Smart Cities. Science for the City. Fortaleza, Brazil. October 2013

Using Swarm Intelligence and Network Sciences in Crime Modeling. Round table at BRICS-CCI. Recife, Brazil. September 2013

Evaluating the Performance of Social Networks of Sensors under Different Mobility Models. University of Fortaleza, Brazil. September 2013

Advances in Network Science fundamental research. How are we doing? Panel Discussion. IEEE Network Science Workshop. May 2013

Short Course on Network Science. AGH University of Science and Technology. Krakow, Poland. June 2012

Research at BioComplex Lab, Florida Tech. State University of Pernambuco. May 2011

Short Course on Complex Networks. University of Fortaleza, Brazil. August 2010

Short Course on Bio-Inspired Computing. University of Fortaleza, Brazil. June 2006

Trends in Information Technology. Unifor World. University of Fortaleza. October 2005

What Swarms can Teach Computer Scientists. University of Fortaleza, Brazil. April 2005

STUDENT SUPERVISION

CURRENT PHD STUDENTS

Guilherme Machado: Causality in Human Mobility

University of Exeter

2025 – Present

Andrew Schendl: Using Mobility to Model Access to Greenspaces

University of Exeter

2023 – Present

Julio Taveira: Using Network Science to Model Interstate Road Access in Brazil

University of Exeter

2023 – Present

Guilherme Machado: Inequality in Social Media

University of Exeter

2022 – Present

SELECTED PAST PHD STUDENTS

Liam Berrisford: Estimating Pollution from Secondary Public Datasets

University of Exeter

2020 – 2024

Currently a Research Software Engineer at the University of Exeter.

Sima Farokhnejad: Modelling of Uncertainties in Cattle Epidemics

University of Exeter

2020 – 2024

Currently a Post-Doc at Southampton University, UK.

Ana Jaramillo: Echo Chambers in Science

University of Exeter

2019 – 2023

Currently a post-doc at Technische Universität Graz, Austria.

Clodomir Santana: Using Network Science to Characterise Swarm-based Algorithms

University of Exeter

2019 – 2023

Currently a post-doc at UC Davis, USA.

Josemar F. Cruz: Social groups characterization and dynamics: a network science approach

Florida Institute of Technology

2014 – 2020

Currently a Data Scientist in Canada.

Mahdi Seyednezhad: A Network Driven Approach for Characterizing Emoji Usage in Social Media

Florida Institute of Technology

2015 – 2020

Currently a Senior Data Scientist in the Chicago Region, USA.

Diego P. Silva: A Comprehensive Data-Driven Characterization of Organ Transplantation

Florida Institute of Technology

2014 – 2018

Currently Assistant Professor at Universidade Católica de Pernambuco, Brazil.

Diogo Pacheco: Information Densification of Social Constructs via Behavior Analysis of Social Media Users

Florida Institute of Technology

2013 – 2017

Currently a Senior Lecturer at the University of Exeter, UK.

Basim Mahmood: Social Capital in Wireless Sensor Networks*Florida Institute of Technology*

2012 – 2015

Currently an Associate Professor at the University of Mosul.

Hugo Barbosa-Filho: Frequency and Recency of Visits in Human Mobility*Florida Institute of Technology*

2012 – 2016

Currently a Lecturer in Urban Systems, University of Exeter, UK.

SERVICE ACTIVITIES

STEERING COMMITTEES AND BOARDS

Board Member, Latin American Conference on Complex Networks	2017 – Present
Co-Editor-in-Chief, Applied Network Science Journal, Springer Verlag	2015 – Present
Board Member, Complexity Series, Springer Verlag	2016 – Present
Board Member, Complex Systems Society	2023 – Present
Associate Editor, Complexity Journal	2018 – Present
Editor, PLoS One	2020 – Present
Associate Editor, Entropy Journal	2024 – Present
Associate Editor, Complexities	2025 – Present
Co-Chair, Conference on Complex Networks (CompleNet)	2009 – 2025
Associate Editor, Learning & Nonlinear Models	2016 – 2024
Board Member, Network Science Society	2013 – 2025
IEEE Task Force on Network Science	2011 – 2013
ACM SIGAPP: Special Interest Group on Applied Computing	2003 – 2009
International Journal of Aviation Technology, Engineering and Management	2011 – 2013

RECENT CONFERENCE ORGANISATION

General Co-Chair. LANET 2027. Porto Alegre, Brazil.	July 2027
General Co-Chair. CompleNet 2025. Fortaleza, Brazil.	April 2025
Program Co-Chair. Conference on Complex Systems. Exeter, UK.	September 2024
General Co-Chair. CompleNet 2024. Exeter, UK.	April 2024
Organising Committee Member. NetSci-X. Venice, Italy.	2024
Organising Committee Member. Conference on Complex Systems (CCS 2023). Salvador, Brazil. October 2023	
Co-Chair. CompleNet 2023. Aveiro, Portugal.	April 2023
Co-Chair. GoodIT. Limassol, Cyprus.	2022
General Chair. CompleNet Live. Exeter, UK.	2020 – 2022
General Chair. CompleNet. Exeter, UK.	2020

Program Chair. IDEAL. Manchester.	2019
Organising Committee Member. NetSci-X. Wroclaw, Poland.	2016
Chair. NetSci-X. Rio de Janeiro.	2015
Chair. Workshop on Complex Networks (CompleNet).	2009 – Present

REFERENCES

Available upon request.